Mini-project 2 : Contrôle d’accès et Cryptographie

Partie 2A : configuration des permissions sur un système linux

1. création de user et groupe

```
┌──(lea㉿kali)-[~]
└─$ sudo adduser jade
[sudo] Mot de passe de lea :
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd : mot de passe mis à jour avec succès
Modifier les informations associées à un utilisateur pour jade
Entrer la nouvelle valeur, ou appuyer sur ENTER pour la valeur par défaut
        NOM []:
        Numéro de chambre []:
        Téléphone professionnel []:
        Téléphone personnel []:
        Autre []:
Is the information correct? [Y/n]

┌──(lea㉿kali)-[~]
└─$ sudo addgroup students

┌──(lea㉿kali)-[~]
└─$ sudo usermod -aG students jade
```

2. création de fichiers et répertoires de test :

```
┌──(lea㉿kali)-[~]
└─$ mkdir test_dir

┌──(lea㉿kali)-[~]
└─$ touch test_file.txt

┌──(lea㉿kali)-[~]
└─$ echo "test test" > test_file.txt

┌──(lea㉿kali)-[~]
└─$ cat test_file.txt
test test
```

3. affichage des permissions actuelles

```
┌──(lea㉿kali)-[~]
└─$ ls -l test_file.txt
-rw-rw-r-- 1 lea lea 10 15 avril 23:20 test_file.txt

┌──(lea㉿kali)-[~]
└─$ ls -l test_dir
total 0
```

4. modification des permissions avec chmod

```
┌──(lea㉿kali)-[~]
└─$ chmod u+x test_file.txt

┌──(lea㉿kali)-[~]
└─$ chmod g-w test_dir

┌──(lea㉿kali)-[~]
└─$ chmod 755 test_dir

┌──(lea㉿kali)-[~]
└─$ chmod 640 test_file.txt

┌──(lea㉿kali)-[~]
└─$ ls -l test_file.txt
-rw-r----- 1 lea lea 10 15 avril 23:20 test_file.txt

┌──(lea㉿kali)-[~]
└─$ chmod u+r test_file.txt

┌──(lea㉿kali)-[~]
└─$ chmod o-r test_file.txt

┌──(lea㉿kali)-[~]
└─$ chmod 670 test_dir
```

modification de droits

5. changement de propriétaire et groupe avec chgrp et chown

```
┌──(lea㉿kali)-[~]
└─$ sudo chown jade test_file.txt

┌──(lea㉿kali)-[~]
└─$ sudo chgrp students test_dir

┌──(lea㉿kali)-[~]
└─$ sudo chown jade:students test_file.txt
```

```
drw-rwx---  2 lea  students   4096 15 avril 23:19 test_dir
-rw-r-----  1 jade students     10 15 avril 23:20 test_file.txt
```

Les modifications ont bien été appliquées.

Partie 2B : mise en place d'un chiffrement basique avec OpenSSL

1. chiffrement d'un fichier avec openssl

```
┌──(lea㉿kali)-[~]
└─$ openssl enc -aes-256-cbc -e -in secret.txt -out secret.enc -k "ma_cle"
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

┌──(lea㉿kali)-[~]
└─$ openssl enc -aes-256-gcm -e -in secret.txt -out secret.enc -k "ma_cle"
enc: AEAD ciphers not supported
enc: Use -help for summary.
```

chiffrement avec deux algos différents.

2. visualisation du fichier chiffré

```
┌──(lea㉿kali)-[~]
└─$ cat secret.enc
Salted__◆
◆`@ж◆j◆yæ◆%◆◆◆◆◆!R◆◆◆◆◆◆ ~◆◆    ◆/◆o◆:
```

non accessible car chiffré.

3. déchiffrement avec openssl

```
┌──(lea㉿kali)-[~]
└─$ openssl enc -aes-256-cbc -d -in secret.enc -out secret_dechiffre.txt -k "ma_cle"
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

┌──(lea㉿kali)-[~]
└─$ cat secret.enc
Salted__◆
◆`@ж◆j◆yæ◆%◆◆◆◆◆!R◆◆◆◆◆◆ ~◆◆    ◆/◆o◆:

┌──(lea㉿kali)-[~]
└─$ openssl enc -aes-256-gcm -d -in secret.enc -out secret_dechiffre.txt -k "ma_cle"
enc: AEAD ciphers not supported
enc: Use -help for summary.

┌──(lea㉿kali)-[~]
└─$ cat secret_dechiffre.txt
message confidentiel attention
```

on déchiffre le message !

Partie 2C : création et manipulation de certificats auto-signés
1. Génération d'une Clé Privée:

```
┌──(lea㉿kali)-[~]
└─$ openssl genrsa -out cle_privee.pem 2048
```

2. Génération d'une Requête de Signature de Certificat (CSR) :

```
┌──(lea㉿kali)-[~]
└─$ openssl req -new -key cle_privee.pem -out requete.csr

You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:FR
State or Province Name (full name) [Some-State]:
Locality Name (eg, city) []:
Organization Name (eg, company) [Internet Widgits Pty Ltd]:ULU
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:ULU
Email Address []:ULU@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:password
An optional company name []:
```

3. Génération du Certificat Auto-Signé :

```
┌──(lea㉿kali)-[~]
└─$  openssl x509 -req -in requete.csr -signkey cle_privee.pem -out certificat.crt -days 365
Certificate request self-signature ok
subject=C=FR, ST=Some-State, O=ULU, CN=ULU, emailAddress=ULU@gmail.com
```

4. Visualisation du Certificat :

```
┌──(lea㉿kali)-[~]
└─$ openssl x509 -in certificat.crt -text -noout

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            50:60:b4:c3:c2:1e:9f:6c:69:64:8c:34:7d:01:66:f5:5c:97:c1:df
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=FR, ST=Some-State, O=ULU, CN=ULU, emailAddress=ULU@gmail.com
        Validity
            Not Before: Apr 17 13:57:16 2026 GMT
            Not After : Apr 17 13:57:16 2027 GMT
        Subject: C=FR, ST=Some-State, O=ULU, CN=ULU, emailAddress=ULU@gmail.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:cc:37:44:14:95:19:28:11:6d:a4:90:b9:e9:8d:
                    0d:78:2c:31:a5:97:b1:8b:62:aa:f2:ec:be:8d:9d:
                    2d:8e:83:2e:5d:7e:08:cc:eb:5e:92:68:86:a4:12:
                    c0:6f:68:d3:a0:83:8f:71:92:6b:da:0d:10:cc:da:
                    47:9f:a3:80:42:6e:7d:4a:e5:4d:b6:f6:c0:33:02:
                    5a:a0:42:62:0f:d7:26:b6:0a:7a:13:0a:53:43:7d:
                    56:96:8e:d2:8b:14:20:4b:fd:47:98:fb:af:77:c3:
                    87:4a:07:2d:7a:51:d3:6d:86:df:9b:d1:ab:d3:08:
                    36:32:51:09:71:cf:ee:7a:01:2b:b3:b1:be:76:b5:
                    81:78:27:06:5a:7b:68:90:7a:0f:f9:a5:33:b1:dd:
                    71:2d:17:f8:57:d2:fb:13:2c:6e:80:40:64:a4:df:
                    95:53:e3:c0:b2:a8:c2:39:43:22:71:4d:77:ed:a4:
                    f0:cd:c0:04:ba:13:2a:8f:2a:3d:b6:bc:19:c9:66:
                    14:26:79:1f:86:47:0e:af:f0:c4:21:7a:2f:7a:93:
                    39:7a:11:c4:3e:dc:ff:36:74:4f:db:58:ec:eb:12:
                    eb:5f:27:07:78:5e:c6:fd:b8:9c:4b:03:8e:3b:94:
                    29:af:03:66:13:49:ae:5a:98:fe:8b:a4:83:8c:a1:
                    73:6f
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                07:11:8B:38:BB:00:6D:23:76:FC:99:A3:11:49:14:63:A6:68:7A:5F
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        a4:38:41:5f:fc:a9:24:63:85:b2:d3:50:36:b4:d2:a8:e5:08:
        1b:23:e3:52:ef:91:54:8a:41:a9:75:85:cb:f1:4a:87:59:c7:
        58:08:38:1d:66:f5:77:c0:b0:03:ce:a0:d0:2f:cf:c8:45:7e:
```

on voit bien toutes les informations du certificat.

5. Manipulation des Certificats:

```
┌──(lea㉿kali)-[~]
└─$ openssl x509 -in certificat.crt -outform DER -out certificat.der


┌──(lea㉿kali)-[~]
└─$ openssl x509 -in certificat.crt -pubkey -noout > cle_publique.pem


┌──(lea㉿kali)-[~]
└─$ cat cle_publique.pem
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAzDdEFJUZKBFtpJC56Y0N
eCwxpZexi2Kq8uy+jZ0tjoMuXX4IzOtekmiGpBLAb2jToIOPcZJr2g0QzNpHn6OA
Qm59SuVNtvbAMwJaoEJiD9cmtgp6EwpTQ31Wlo7SixQgS/1HmPuvd8OHSgctelHT
bYbfm9Gr0wg2MlEJcc/uegErs7G+drWBeCcGWntokHoP+aUzsd1xLRf4V9L7Eyxu
gEBkpN+VU+PAsqjCOUMicU137aTwzcAEuhMqjyo9trwZyWYUJnkfhkcOr/DEIXov
epM5ehHEPtz/NnRP21js6xLrXycHeF7G/bicSwOOO5QprwNmE0muWpj+i6SDjKFz
bwIDAQAB
-----END PUBLIC KEY-----
```

création d'un fichier avec clé publique

Partie 2D : Mise en place d'un VPN (OpenVPN).

1. Installation d'OpenVPN:

```
┌──(lea㉿kali)-[~]
└─$ sudo apt install openvpn easy-rsa
openvpn est déjà la version la plus récente (2.7.0-1).
Le paquet suivant a été installé automatiquement et n'est plus nécessaire :
  libjs-underscore
Veuillez utiliser « sudo apt autoremove » pour le supprimer.

Mis à jour :
  easy-rsa

Sommaire :
  Mise à niveau de : 1. Installation de : 0Supprimé : 0. Non mis à jour : 2105
Taille du téléchargement : 82,0 kB
  Espace nécessaire : 4 096 B / 38,1 GB disponible

Continuer ? [O/n] o
Réception de : 1 http://kali.download/kali kali-rolling/main amd64 easy-rsa all 3.2.6-1 [82,0 kB]
82,0 ko réceptionnés en 4s (21,9 ko/s)
(Lecture de la base de données ... 416909 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../easy-rsa_3.2.6-1_all.deb ...
Dépaquetage de easy-rsa (3.2.6-1) sur (3.2.5-1) ...
Paramétrage de easy-rsa (3.2.6-1) ...
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
Traitement des actions différées (« triggers ») pour kali-menu (2025.2.7) ...
```

2. Configuration du Serveur OpenVPN:

```
┌──(lea㉿kali)-[~]
└─$ sudo mkdir /etc/openvpn/server
sudo cp /usr/share/doc/openvpn/examples/sample-config-files/server.conf
/etc/openvpn/server/server.conf
mkdir: cannot create directory '/etc/openvpn/server': Le fichier existe
cp: opérande de fichier cible manquant après '/usr/share/doc/openvpn/examples/sample-config-files/server.conf'
Saisissez « cp --help » pour plus d'informations.
zsh: permission non accordée: /etc/openvpn/server/server.conf

┌──(lea㉿kali)-[~]
└─$ sudo cp /usr/share/doc/openvpn/examples/sample-config-files/server.conf /etc/openvpn/server/server.conf
[sudo] Mot de passe de lea :
```

```
# DNS servers provided by opendns.com.
;push "dhcp-option DNS 208.67.222.222"
;push "dhcp-option DNS 208.67.220.220"
push "dhcp-option DNS 8.8.8.8"
# Uncomment this directive to allow different
```

```
# TCP or UDP server?
;proto tcp
proto udp
```

on utilise un serveur udp

```
ca ca.crt
cert server.crt
key server.key  # This file should be kept secret
```

3. Génération des Clés et des Certificats pour le Serveur:

```
┌──(lea㉿kali)-[~]
└─$ sudo make-cadir /etc/openvpn/easy-rsa

/etc/openvpn/easy-rsa exists. Aborting.
Usage: /usr/bin/make-cadir DIRECTORY
Creates a *new* directory and prepares it to be used as a (CA) key management directory (to create and store keys a
nd certificates).
```

```
┌──(root㉿kali)-[/home/lea]
└─# cd /etc/openvpn/easy-rsa

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# sudo ./easyrsa init-pki


Notice
------

'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /etc/openvpn/easy-rsa/pki

IMPORTANT: PKI algorithm is

Using Easy-RSA Auto-configured 'vars' file:
* /etc/openvpn/easy-rsa/vars


┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# sudo ./easyrsa build-ca nopass

Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
.+.....+ ... ++++++++++++++++++++++++++++++++++++++++*.......+.....+ ... +.+ .. + ... +.+........+............+....+.....+.
.....+ ... +.+............+ .. +......+....+ .. +......+ ... +.......+ ... + .. +....+........+.++++++++++++++++++++++++++++++++
++++++++*....+.........+ ... +........+.........+............+ .. + ... + ... +.......+ .. +.....+.....+.........+ ... + ... +.+.
............+.......+.....+ ... +.......+........+....+......+.........+ ... + .. +...............+......+.........+ ... +
 ... +.+......+ .. +...............+.......+.....+......+.........+.++++++
 ... +.....++++++++++++++++++++++++++++++++++++++++*.....+ ... +......................++++++++++++++++++++++++++++++++++
++++++*.+.......+ ... +...........+...............+....+ .. + ... +......+..........+ .. +...........+.....+...........+......
 .. +....+.....+.......+..............+.+ ... + ... +.....+ ... +.+ ... +...........+.+.....+.+ .. +......+.+.....+.+.....+....
+...........+ ... +....+............+.....+....+ .. +....................+.....+ ... +......+.........+......+....+ ... + .. +
......+....+ .. +....................+..........+.....+.+.........+............+.....+.+ .. +.+.....+....................
.+.........+.........+ .. +.........+......+.........+....+ .. +.......+ ... +.....+....+ ... +...........+..........+....
 .. + ... + ... + .. +......+.+.....+......+.......+....................+.+....................+.+.........................
.+...............+ ... +.+.....+.........+....+...........................+.+.......+ .. + ... +.......+ .. +.........+......
............+.....+......+.+ ... + ... +........+.........+................+.....++++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:

Notice
```

passage en sudo

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─#  sudo ./easyrsa build-server-full server nopass

Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
 .. + ... +.....+................+.....+.+...........+....+.....+......+....+......+........+....+................+ .. + ..
 .+......++++++++++++++++++++++++++++++++++++++++++*........+.............+ .. + ... +.......+.....+.........+ ... +.........
 .+............+......+ ... ++++++++++++++++++++++++++++++++++++++++++* .. +.+........+.+...........................+....+ ...
 .. + ... +.+.........+........+.........+ .. +.......+........+ ... +.......+ ... +.....+.......+.....+ .. + ... +............
 .........+....+.........+.....+ ... +.........+ ... +.............+.....+ ... +.....................+....+.....+....+ .. +.
 .....+.......+.....+......+.+........+......+.+ ... + .. +.+........+.........+.........+...........+.+...........+ ...
 ......+ ... +.+......+.........+......+ ... +.....+......+.......+ ... +......+........+......+....+................+ ... +.
 .. + .. + ... +....+............+ ... +.....+.....+....+ ... +..............+.+................+ .. +.......+.............+ ..
 ....+....+...........+....+ .. +....+.....+...........+ ... +.......+ .. +....................+.....+.............+ .. +......
 .......+ .. +.........+......+ ... +.+..............+.......+......+ ... +.........+............+......+ .. +.+.....+.+....
 .+...........+ ... + ... +.....+........+........+....+ ... +............+.+ ... +.....................+......+...........+ ...
 .........+.......+.........+.......................+......+ .. + ... +.+......+.....+ ... +.+......+ .. +......+.+......+.
 .+.+...........++++++
 ... +......+.....+....+++++++++++++++++++++++++++++++++++++++++* ... + ... + ... +......++++++++++++++++++++++++++++++++++++
```

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# sudo openssl dhparam -out dh.pem 2048

Generating DH parameters, 2048 bit long safe prime
.....................................................................................+.....
.....................................................................+.....................
.....................................................................................
.....................................................................................
.....................................+.................................+.....................
....+................................................................................
.....................................................................++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++*
```

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# sudo cp pki/ca.crt /etc/openvpn/server
sudo cp pki/issued/server.crt /etc/openvpn/server
sudo cp pki/private/server.key /etc/openvpn/server
sudo cp dh.pem /etc/openvpn/server

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# sudo chown -R root:root /etc/openvpn/easy-rsa


┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─#
```

4. Configuration du Client VPN:

```
┌──(kali㉿kali)-[~/easy-rsa]
└─$ ./easyrsa build-ca
Using Easy-RSA 'vars' configuration:
* /home/kali/easy-rsa/vars

Enter New CA Key Passphrase:
Passphrase must be at least 4 characters!

Enter New CA Key Passphrase:

Confirm New CA Key Passphrase:
.....+..........+......+ ... + .. +.........+ ... + ... ++++++++++++++++++++++++++++++++++++++++*.......+ ... +......+.+ ... + ..
.+ .. +.......+..............+......+.+ ... + .. ++++++++++++++++++++++++++++++++++++++++*....+.+ .. +......+.+......+ .. + ...
.........+.........+....+ ... +.....+.......+ .. +........................+.......+.....+....+.....+................+ ..
.+......+.....+......+ ... +....+ .. +..................+..................+ ... +.+........+.+.....+.......+ ... +.....+ ..
.......+ ... +.........+.+..............+.+.....+............+..........+......+......+............+ ... +......+....+ ..
..........+ .. + ... +.......+........+....+.........+.....+....+ ... +.....+.......+ .. +......+ ... +....+......+.....+....
.. +..........+.....+.+ .. +.+......+..............+.+..............+....+........+.......+ .. +......+ ... +.+...........
.........+.+ ... +............+ ... +............+........+....+ .. +.........+......+.+ ... +......................+.......
....+ ... +....+ .. +....+......+.....+.+.....+ ... +....+ .. +......+.......+........+.......+........+.........+......+ ..
.+.+......+ ... ++++++
.+ ... +.......+.....+ ... +....+......+.........+ ... + .. +..........+........+.+ .. +.........+ ... +......+................+
++++++++++++++++++++++++++++++++++++++*.....+.++++++++++++++++++++++++++++++++++++++++*.. +.........++++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:

Notice
------
CA creation complete. Your new CA certificate is at:
* /home/kali/easy-rsa/pki/ca.crt

Build-ca completed successfully.


┌──(kali㉿kali)-[~/easy-rsa]
└─$ ./easyrsa build-client-full client1 nopass
Using Easy-RSA 'vars' configuration:
* /home/kali/easy-rsa/vars
```

on refait le setup de CA et initialisation du pki du client

```
Fichier  Actions  Éditer  Vue  Aide
  GNU nano 8.4                                        ca.crt
  GNU nano 8.4
-----BEGIN CERTIFICATE-----
MIIDTjCCAjagAwIBAgIUHYGs5/M7RIPrreB+O7Kzran0UDMwDQYJKoZIhvcNAQEL
BQAwFjEUMBIGA1UEAwwLRWFzeS1SU0EgQ0EwHhcNMjYwNDE3MTYzNzU4WhcNMzYw
NDE0MTYzNzU4WjAWMRQwEgYDVQQDDAtFYXN5LVJTQSBDQTCCASIwDQYJKoZIhvcN
AQEBBQADggEPADCCAQoCggEBAIN7KR0h7jkBKwYs0MM8fojsPsN6j/bKWxU/ebRx
gsm+x25oDYWtXvjY4TgLGeRlzuvdOQLvh7DZQ4lv9VUGW+e4ToE+PzuxUie6pP02
BSOCA0UL4mqy6jAE8LZ7gge2btldkTNxycMdJmrGy2OgB57qE1dorgKKlraZRdN6
RMShg1UeahGOTPVF+wEsHRrpe7TcCRvJHeM6mfw4+pHsE+ID7Ym/FbSZnMlTzaET
06tyLCmc51C+t5Vo9MKQ7Ba51ICU3+7/U7Tx228B70REl4VQJnLYjvZF6/ntutbw
DlTYcvw1wUENCnKkypt5oNgKRZiEGm4idwhwHxiN5SKuhxsCAwEAAaOBkzCBkDAP
BgNVHRMBAf8EBTADAQH/MB0GA1UdDgQWBBTrCes3ykmGsslK3FfRSXWC0eqztjBR
BgNVHSMESjBIgBTrCes3ykmGsslK3FfRSXWC0eqztqEapBgwFjEUMBIGA1UEAwwL
RWFzeS1SU0EgQ0GCFB2BrOfzO0SD663gfjuys62p9FAzMAsGA1UdDwQEAwIBBjAN
```

copie du ca.crt sur le client

```
┌──(kali㉿kali)-[/etc/openvpn/easy-rsa/pki]
└─$ sudo systemctl status openvpn.service

● openvpn.service - OpenVPN service
     Loaded: loaded (/usr/lib/systemd/system/openvpn.service; disabled; preset: disabled)
     Active: active (exited) since Fri 2026-04-17 18:56:53 CEST; 4s ago
 Invocation: f986ea933a05417bab264e46738e1acf
       Docs: man:openvpn(8)
    Process: 10063 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 10063 (code=exited, status=0/SUCCESS)
   Mem peak: 1.8M
        CPU: 16ms
```

on start le service openvpn

6.connexion du client openvpn

```
┌──(kali㉿kali)-[~]
└─$ sudo openvpn --config client.ovpn
2026-04-17 19:12:21 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.

2026-04-17 19:12:21 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cip
her negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your
configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers DEFAULT:BF-CBC
2026-04-17 19:12:21 Note: Kernel support for ovpn-dco missing, disabling data channel offload.
2026-04-17 19:12:21 OpenVPN 2.7.0 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AE
AD] [DCO]
2026-04-17 19:12:21 library versions: OpenSSL 3.5.0 8 Apr 2025, LZO 2.10
2026-04-17 19:12:21 DCO version: N/A
2026-04-17 19:12:21 WARNING: No server certificate verification method has been enabled.  See http://openvpn.net/ho
wto.html#mitm for more info.
2026-04-17 19:12:21 TCP/UDP: Preserving recently used remote address: [AF_INET]192.168.56.101:1194
2026-04-17 19:12:21 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-04-17 19:12:21 UDPv4 link local: (not bound)
2026-04-17 19:12:21 UDPv4 link remote: [AF_INET]192.168.56.101:1194
```

```
┌──(kali㉿kali)-[~/easy-rsa]
└─$ sudo openvpn --config /etc/openvpn/server/server.conf
2026-04-17 19:10:59 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.

2026-04-17 19:10:59 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cip
her negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your
configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers DEFAULT:BF-CBC
2026-04-17 19:10:59 Note: Kernel support for ovpn-dco missing, disabling data channel offload.
2026-04-17 19:10:59 OpenVPN 2.7.0 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AE
AD] [DCO]
2026-04-17 19:10:59 library versions: OpenSSL 3.5.0 8 Apr 2025, LZO 2.10
2026-04-17 19:10:59 DCO version: N/A
2026-04-17 19:10:59 net_route_v4_best_gw query: dst 0.0.0.0
2026-04-17 19:10:59 net_route_v4_best_gw result: via 192.168.1.254 dev eth0
2026-04-17 19:10:59 NOTE: your local LAN uses the extremely common subnet address 192.168.0.x or 192.168.1.x.  Be a
ware that this might create routing conflicts if you connect to the VPN server from public locations such as intern
et cafes that use the same subnet.
2026-04-17 19:10:59 Diffie-Hellman initialized with 2048 bit key
2026-04-17 19:10:59 TUN/TAP device tun0 opened
2026-04-17 19:10:59 tun/tap device [tun0] opened
2026-04-17 19:10:59 net_iface_mtu_set: mtu 1500 for tun0
2026-04-17 19:10:59 net_iface_up: set tun0 up
2026-04-17 19:10:59 net_addr_v4_add: 10.8.0.1/24 dev tun0
2026-04-17 19:10:59 Could not determine IPv4/IPv6 protocol. Using AF_INET
2026-04-17 19:10:59 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-04-17 19:10:59 UDPv4 link local (bound): [AF_INET][undef]:1194
2026-04-17 19:10:59 UDPv4 link remote: [AF_UNSPEC]
2026-04-17 19:10:59 MULTI: multi_init called, r=256 v=256
2026-04-17 19:10:59 IFCONFIG POOL IPv4: base=10.8.0.2 size=253
2026-04-17 19:10:59 MULTI IO: MULTI_IO INIT maxclients=1024 maxevents=1029
2026-04-17 19:10:59 Initialization Sequence Completed
```

```
kali@kali: ~
Fichier  Actions  Éditer  Vue  Aide
    link/ether 08:00:27:23:7f:0c brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.149/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
       valid_lft 84671sec preferred_lft 84671sec
    inet6 2001:861:36c7:9c50:b506:15dd:2388:2272/64 scope global temporary dynamic
       valid_lft 86365sec preferred_lft 14365sec
    inet6 2001:861:36c7:9c50:a00:27ff:fe23:7f0c/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86365sec preferred_lft 14365sec
    inet6 fe80::a00:27ff:fe23:7f0c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.1/24 brd 10.8.0.255 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::a677:8ea6:a906:a79e/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
```

En lançant le fichier de conf on réussit bien à se connecter au vpn !